

Examination Repeat & Medical Application Management System (ERMAS)

Software Architecture, Technology Stack and Development Roadmap

Version 1.0

1. Executive Summary

The Examination Repeat & Medical Application Management System (ERMAS) is a centralized web-based application designed to automate and digitize the management of Repeat and Medical examination applications.

The system replaces the existing manual paper-based process with a secure, role-based workflow platform that supports:

- Student application submission
 - Payment verification
 - Examination verification
 - Multi-level approvals
 - Examination schedule management
 - Reporting and analytics
 - Audit logging
 - Document management
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2. Project Objectives

Primary Objectives

- Eliminate paper applications.
 - Reduce manual processing.
 - Improve transparency.
 - Enable application tracking.
 - Improve data security.
 - Provide management reports.
 - Support future expansion.
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3. System Scope

The system shall support:

- Applied Accounting Degree Programme
- BMBA Degree Programme

The architecture shall support future degree programmes without system redesign.

4. User Roles

Role	Responsibility
Student	Submit applications
Finance Officer	Verify payments
Schedule Officer	Manage examination schedules
Verification Officer	Verify eligibility
Exam Manager	Examination approval
Registrar	Final approval
Director	Exceptional approvals
Super Admin	Full system administration

5. Authentication

Students

Login Method:

- Batch Number
- NIC Number

Example:

Batch No: AA22-105

NIC: 200012345678

Staff Members

Login Method:

- Email address
- Password

Optional:

- Two-factor authentication
 - Password reset
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6. Permission-Based Access Control

The system shall implement a Permission-Based Access Control (PBAC) model.

Structure:

User ↓ Roles ↓ Permissions

Users may have multiple roles and permissions.

Example:

User:

- Finance Officer
- Schedule Officer

Permissions:

- Verify Payments
 - Create Schedules
 - Edit Schedules
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7. Core Modules

Student Portal

- Submit applications
- Upload documents
- Upload payment slips

- View application status
 - View history
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Examination Management

- Examination periods
 - Examination schedules
 - Subject management
 - Semester management
-

Finance Management

- Payment verification
 - Payment reports
-

Application Management

- Repeat applications
 - Medical applications
 - Verification process
-

Workflow Management

- Multi-stage approvals
 - Remarks
 - Returns
 - Rejections
-

Administration

- User management
 - Role management
 - Permission management
 - Audit logs
-

8. Technology Stack

Frontend

- Next.js 15
 - React 19
 - TypeScript
 - Tailwind CSS
 - shadcn/ui
 - TanStack Table
 - React Hook Form
 - Zod Validation
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Backend

- NestJS
 - TypeScript
 - Prisma ORM
 - JWT Authentication
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Database

- PostgreSQL
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Cache and Queue

- Redis
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File Storage

- MinIO
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Infrastructure

- Docker
- Nginx
- Ubuntu Server
- Cloudflare

9. System Architecture

Internet ↓ Cloudflare ↓ Nginx Reverse Proxy ↓ Next.js Frontend ↓ NestJS API ↓

Authentication Service Application Service Finance Service Workflow Service Notification Service Reporting Service

↓ PostgreSQL Redis MinIO

10. Application Workflow

Student ↓ Finance Verification ↓ Verification Officer ↓ Exam Manager ↓ Registrar ↓ Director (Optional) ↓ Completed

Actions:

- Approve
 - Reject
 - Return
 - Escalate
-

11. Database Modules

User Management

- users
 - roles
 - permissions
 - user_roles
 - role_permissions
-

Academic Management

- programmes
- batches
- students

- subjects
-

Examination Management

- exam_periods
 - exam_schedules
-

Application Management

- applications
 - application_subjects
 - documents
 - payments
-

Workflow Management

- approval_logs
 - remarks
-

System Management

- notifications
 - audit_logs
-

12. Security Architecture

Authentication

- JWT Access Token
- Refresh Tokens

Password Security

- Argon2 Password Hashing

Security Features

- Rate limiting

- CSRF protection
 - XSS protection
 - File validation
 - Session management
 - Activity logging
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13. Document Management

Supported Formats:

- PDF
- JPG
- PNG

Document Types:

- Medical certificates
- Payment slips
- Supporting documents

Storage Technology:

- MinIO Object Storage
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14. Notification System

Notifications:

- Application submitted
- Payment verified
- Application approved
- Application rejected
- Returned for correction

Channels:

- In-system notifications
 - Email notifications (future)
-

15. Reporting Module

Reports:

- Applications by programme
- Applications by batch
- Payment reports
- Revenue reports
- Pending approvals
- User activities

Export Formats:

- PDF
 - Excel
-

16. User Interface Structure

Dashboard

Applications

- Repeat Applications
- Medical Applications

Examinations

- Examination Periods
- Schedules

Finance

Reports

Administration

Settings

17. Development Roadmap

Phase 1 – Project Foundation

Duration: 2 Weeks

Tasks:

- Project setup
 - Database design
 - Authentication
 - User management
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Phase 2 – Student Portal

Duration: 2 Weeks

Tasks:

- Student login
 - Application forms
 - File uploads
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Phase 3 – Finance Module

Duration: 1 Week

Tasks:

- Payment verification
 - Financial reports
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Phase 4 – Examination Module

Duration: 2 Weeks

Tasks:

- Examination schedules
- Examination periods

Phase 5 – Workflow Module

Duration: 2 Weeks

Tasks:

- Approval workflow
- Remarks
- Return process

Phase 6 – Reporting Module

Duration: 1 Week

Tasks:

- Dashboard reports
- Export functionality

Phase 7 – Testing and Security

Duration: 1 Week

Tasks:

- Security testing
- Performance testing
- User acceptance testing

18. Deployment Architecture

Internet ↓ Cloudflare ↓ Nginx ↓ Docker Containers

Frontend Container Backend Container Redis Container PostgreSQL Container MinIO Container

19. Future Enhancements

- Online payment gateway
 - SMS notifications
 - Microsoft 365 integration
 - Digital signatures
 - QR verification
 - Mobile application
 - AI document verification
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20. Estimated Project Timeline

Phase	Duration
Analysis & Design	1 Week
Development	8 Weeks
Testing	1 Week
Deployment	1 Week

Total Estimated Duration:

10–12 Weeks

21. Repository Structure

exam-system/

apps/

- web
- api

packages/

- ui
- auth
- shared

infrastructure/

- docker

- nginx

docs/

Conclusion

ERMAS provides a secure, scalable, and modern examination application management platform that replaces manual processes and supports future institutional growth.

The architecture is designed to support:

- Multiple degree programmes
- Additional workflows
- Future integrations
- Increased user load
- Long-term maintainability

Version: 1.0 Prepared by: Solution Architecture & System Design Team